

# RxChat ? RAG Chatbot Implementation Plan

> Production architecture and implementation roadmap for RxChat's Retrieval-Augmented Generation pipeline.

> For detailed explanations of technical concepts, see the companion [Explanation Document](./CHATBOT\_PLAN\_EXPLAINED.md).

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## Stack Overview

| Component | Technology | Purpose |

|:---|:---|:---|

| **Frontend** | React + Vite ? Vercel | Chat UI, auth flows, pre-chat disclaimer |

|                    |                       |                                        |
|--------------------|-----------------------|----------------------------------------|
| <b>**Backend**</b> | Django + DRF ? Render | API, RAG orchestrator, background jobs |
|--------------------|-----------------------|----------------------------------------|

**Database** | Supabase Postgres | Users, conversations, messages, audit logs |

**\*\*Vector DB\*\*** | Qdrant Cloud (unified collection) | Drug data embeddings + hybrid search |

**\*\*Embeddings\*\*** | OpenAI `text-embedding-3-small` | Query + document embedding (1536-dim) |

| Model          | DeepSeek (prefix caching + SSE) | Role-aware answer generation |
|----------------|---------------------------------|------------------------------|
| <b>**LLM**</b> | DeepSeek (prefix caching + SSE) | Role-aware answer generation |

|                       |                             |                                 |
|-----------------------|-----------------------------|---------------------------------|
| <b>**Task Queue**</b> | Django-Q2 (Postgres broker) | Ingestion jobs, scheduled tasks |
|-----------------------|-----------------------------|---------------------------------|

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### ## Architecture Diagram

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[illegible]

? User (React SPA on Vercel) ?

? ??? Chat UI with SSE streaming ?

? ??? Pre-chat medical disclaimer (shown once per session) ?

? ??? Auth: email ? OTP ? profile (role set in profile) ?

????????????????????????????????????????????????????????????????????????????????????

? HTTPS / SSE

????????????????????????????????????????????????????????????????????????????????????

? Django API on Render ?

? ??? Auth endpoints (existing) ?

? ??? Chat endpoints (existing ? RAG injected transparently) ?

? ??? ai\_service.py ? prompt assembly + DeepSeek streaming ?

? ??? rag\_service.py ? embed ? search ? filter ? format ?

? ? ? ?

? ??? 1. Reformulate query (conversation-aware) ?

? ??? 2. Embed query ? OpenAI text-embedding-3-small ?

? ??? 3. Hybrid search ? Qdrant (dense + sparse) ?

? ??? 4. Relevance gate (score ? 0.75 / 0.50 / discard) ?

? ??? 5. Augment prompt ? DeepSeek (stream response) ?

????????????????????????????????????????????????????????????

? ? ?

???????????????? ???? ?????????????

? Supabase ? ?Qdrant ? ? DeepSeek ?

? Postgres ? ?Cloud ? ? API ?

? ? ? ? ? ?

? Users ? ?drug\_ ? ? Prefix ?

? Messages ? ?know- ? ? caching ?

? Audit logs ? ?ledge ? ? SSE ?

? Ingestion ? ?(unified? ? streaming ?

? metadata ? ?collection)? ?

???????????????? ???? ?????????????

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## Data Sources

### Priority & Status

| Priority | Source | Content | Format | License | Status |

|:---|:---|:---|:---|:---|

| **\*\*P0\*\*** | NEML (8th Ed, 2024) | Nigeria essential medicines | PDF | Free (govt) | Download from health.gov.ng |

| **\*\*P0\*\*** | NAFDAC Greenbook | Registered drugs in Nigeria | Web portal | Free (govt) | Scrape from greenbook.nafdac.gov.ng |

| **\*\*P0\*\*** | NHIA STG | Treatment guidelines | PDF (~124 MB) | Free (govt) | Download from nhia.gov.ng |

| **\*\*P1\*\*** | WHO EML | Global essential medicines | Excel | Free | Download from essentialmeds.org |

| **\*\*P1\*\*** | DrugBank | Drug interactions, mechanisms | XML (~700 MB) | CC BY-NC 4.0 | Free for dev; commercial license needed later |

| **\*\*P1\*\*** | OpenFDA Labels | US drug labelling | JSON API | Public domain | Free API, no key needed |

| **\*\*P2\*\*** | EMDEX | Comprehensive Nigerian drug ref | API (licensed) | Commercial | Contact Editor@Emdex.org for licensing |

### Storage Estimates

**\*\*Qdrant Cloud (free tier: 1 GB):\*\***

- P0 sources only: ~55?120 MB ? fits in free tier
- P0 + P1 (curated): ~200?400 MB ?? may need curation
- All sources: ~255?515 MB ?? curate DrugBank/OpenFDA to Nigerian drugs only

**\*\*Supabase Postgres (free tier: 500 MB):\*\***

- Estimated total: ~100?160 MB ? comfortably fits

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**## Retrieval Pipeline**

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User message

?

??? 1. CLASSIFY: drug lookup / general chat / emergency

?     ??? Emergency ? safety escalation (skip RAG)

?

??? 2. REFORMULATE: use conversation context for ambiguous queries

?     ??? "What about side effects?" ? "side effects of metformin"

?

??? 3. EMBED: OpenAI text-embedding-3-small ? 1536-dim vector

?

??? 4. HYBRID SEARCH: Qdrant dense + sparse (BM25)

?     ??? top\_k: 8 (over-retrieve)



? ??? score\_threshold: 0.50 (pre-filter)

? ??? metadata filters: source, category (optional)

?

??? 5. RELEVANCE GATE:

? ??? ? 0.75 ? high-confidence context

? ??? 0.50?0.75 ? supplementary context

? ??? < 0.50 ? discarded

? ??? Zero chunks pass ? NO\_CONTEXT\_NOTE fallback

?

??? 6. PROMPT ASSEMBLY:

? ??? Static prefix (system + behaviour + safety + role)

? ? ??? DeepSeek prefix caching reuses this block

? ??? RAG context (chunks ? NOT shown to user)

? ??? User query

?

??? 7. GENERATE: DeepSeek stream via SSE

?

??? 8. LOG: audit metadata to Postgres

...

### ### Role Handling

- Role is read from `UserProfile` (set during profile setup)
- Retrieval is **role-agnostic** ? same drug data retrieved for all roles
- Generation is **role-aware** ? DeepSeek adjusts tone/depth via role prompts
- If user implies a different role mid-conversation, AI confirms before switching

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## ## Implementation Phases

### ### Phase 1: Infrastructure (Weeks 1-2)

- [ ] Migrate SQLite → Supabase Postgres
- [ ] Create Qdrant Cloud cluster + unified collection `drug\_knowledge`
- [ ] Set up OpenAI API for embeddings
- [ ] Remove `chromadb`, `langchain-\*` from `requirements.txt`
- [ ] Add `qdrant-client`, `pdfplumber` to `requirements.txt`
- [ ] Build PDF parsing infrastructure (for NEML, NAFDAC, NHIA STG)

- [ ] Build base text chunker (1000 chars, 200 overlap)

### ### Phase 2: Ingestion & RAG (Weeks 3?4)

- [ ] Build parsers: NEML, NAFDAC Greenbook, NHIA STG, WHO, DrugBank, OpenFDA
- [ ] Build embedding + upsert pipeline (OpenAI ? Qdrant)
- [ ] Build `rag\_service.py`: embed, hybrid search, relevance gate, reformulation
- [ ] Integrate RAG into `ai\_service.py`
- [ ] Wire into `views.py` `send\_message()` flow
- [ ] Add `IngestionRun` and `AuditLog` models
- [ ] Create `ingest\_drugs` management command
- [ ] Run full ingestion of all P0 + P1 sources

### ### Phase 3: Testing & Deploy (Week 5)

- ☐ Test with 100+ pharmacy queries (Nigeria-specific + general)
- ☐ Test conversation-aware retrieval
- ☐ Test hybrid search accuracy
- ☐ Test role-based response variations
- ☐ Add pre-chat disclaimer to frontend
- ☐ Deploy to Render + Vercel
- ☐ Create golden test set for ongoing evaluation

### ### Phase 4: Hardening (Weeks 6-9)

- ☐ Re-ranking (if hybrid search quality insufficient)

- ☐ Query classification (drug lookup / general / emergency)
- ☐ Rate limiting per subscription tier
- ☐ Prompt injection sanitization
- ☐ Django-Q2 scheduled jobs (data freshness, OTP cleanup, usage)
- ☐ Monitoring + structured logging
- ☐ Weekly RAG evaluation against golden test set
- ☐ EMDEX integration (when licensed)
- ☐ User feedback mechanism (thumbs up/down)

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## ## Key Design Decisions

| Decision | Choice | Rationale |

|:---|:---|:---|

| No source citations in UI | Sources used internally only | User preference ? cleaner chat experience  
|

| Pre-chat disclaimer only | Shown once per session, not per message | User preference ? less  
intrusive |

| Unified Qdrant collection | Metadata handles classification | Simpler ops, flexible filtering |

| Hybrid search from day one | Dense + sparse (BM25) | Drug names need exact keyword matching  
|

| No LangChain | Direct OpenAI + Qdrant clients | Simpler, fewer dependencies |

| Django-Q2 over Celery | Postgres broker | No Redis needed |

| No offline AI fallback | Show error when backend unreachable | User preference ? no fake  
responses |

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## Backend File Changes

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backend/

??? chat/

? ??? ai\_service.py      # [MODIFY] Wire RAG context into prompt

? ??? rag\_service.py      # [NEW] Embed + hybrid search + reformulation

? ??? ingestion/          # [NEW] Data ingestion package

? ? ??? \_\_init\_\_.py

? ? ??? base.py          # Base parser + chunker

? ? ??? neml\_parser.py    # NEML PDF parser

? ? ??? nafdac\_parser.py # NAFDAC Greenbook scraper



? ? ??? nhia\_stg\_parser.py # NHIA STG PDF parser

? ? ??? who\_parser.py # WHO EML Excel parser

? ? ??? drugbank\_parser.py # DrugBank XML parser

? ? ??? openfda\_parser.py # OpenFDA JSON API puller

? ??? management/commands/

? ? ??? ingest\_drugs.py # [NEW] Management command

? ??? models.py # [MODIFY] Add IngestionRun, AuditLog

? ??? views.py # [MODIFY] Wire RAG into send\_message

? ??? tasks.py # [NEW] Django-Q2 task definitions

??? config/settings.py # [MODIFY] Supabase, Qdrant, OpenAI config

??? requirements.txt # [MODIFY] Swap deps

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## Risks & Mitigation

| Risk | Mitigation |

|:---|:---|

| Hallucinated drug info | Relevance threshold, NO\_CONTEXT\_NOTE fallback, pre-chat disclaimer |

| Nigerian PDF parsing quality | Manual QA on parsed chunks; OCR fallback for scanned PDFs |

| DrugBank commercial license | Track source metadata; removable if needed before monetizing |

| EMDEX access | Contact early; defer to P2 if licensing takes time |

| Qdrant free tier overflow | Curate DrugBank/OpenFDA to Nigeria-relevant drugs only |

| DeepSeek data residency | Strip PII before sending; architecture supports LLM swap |